

SHINJO, Japan

WMO: 475200

Lat: 38.7572N

Lon: 140.3120E

Elev: 335

StdP: 14.52

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	20.2	22.7	15.9	12.5	25.9	18.0	13.8	27.9	23.4	30.7	21.1	30.4	4.0	20	0.458

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	15.2	89.8	76.9	87.2	75.5	84.4	73.9	77.9	87.1	76.8	84.9	75.5	82.4	8.4	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.3	134.3	82.3	74.3	129.8	81.2	73.3	125.5	80.1	41.7	87.7	40.5	85.2	39.3	82.6	82.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.2	17.6	15.5	13.4	94.3	3.2	2.5	11.1	96.1	9.2	97.6	7.4	99.0	5.1	100.8
			12.9	79.9	3.2	1.4	10.6	80.9	8.7	81.7	6.9	82.5	4.6	83.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	52.5	30.2	31.1	36.6	47.7	59.4	67.3	74.4	76.7	68.8	56.8	44.5	34.8	(6)
	DBStd	17.15	3.99	4.17	4.58	6.21	5.39	4.44	4.58	4.67	5.79	5.54	5.98	4.69	(7)
	HDD50	2340	614	528	415	117	3	0	0	0	0	9	183	471	(8)
	HDD65	5478	1079	948	880	520	185	26	1	0	26	261	616	936	(9)
	CDD50	3246	0	0	0	47	294	519	758	827	565	219	17	0	(10)
	CDD65	908	0	0	0	0	11	95	293	362	141	6	0	0	(11)
	CDH74	6019	0	0	0	9	162	510	1860	2742	723	13	0	0	(12)
	CDH80	1824	0	0	0	0	23	92	579	968	162	0	0	0	(13)
(14) Wind	WSAvg	6.0	6.7	6.8	6.7	6.7	6.7	6.1	5.5	5.2	5.1	5.0	5.5	6.3	(14)
(15) Precipitation	PrecAvg	75.2	8.3	5.3	4.6	3.8	4.3	5.2	8.3	7.0	5.7	6.1	7.4	9.3	(15)
	PrecMax	86.5	12.6	8.7	7.5	6.1	8.2	11.2	22.6	17.7	11.1	9.4	13.7	17.3	(16)
	PrecMin	58.8	5.3	2.3	2.4	1.1	1.8	1.7	2.6	2.0	2.2	2.9	2.8	3.4	(17)
	PrecStd	6.6	1.8	1.6	1.5	1.5	1.6	2.6	4.1	3.8	2.5	1.8	2.1	2.8	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.4	46.4	57.7	74.8	83.0	86.2	92.4	94.2	88.2	75.4	63.9	52.3	(19)
		MCWB	38.5	41.0	46.1	56.0	63.9	69.2	77.3	78.1	75.5	65.2	56.4	48.9	(20)
	2%	DB	39.4	42.8	51.9	69.0	78.5	82.6	89.0	91.0	84.4	71.5	59.3	47.9	(21)
		MCWB	36.0	37.9	43.8	53.6	61.3	68.2	76.3	77.4	73.9	63.4	53.6	44.4	(22)
	5%	DB	37.5	40.2	47.9	63.9	74.6	79.7	86.2	88.4	81.2	68.8	56.6	44.7	(23)
		MCWB	34.7	35.9	41.4	52.0	59.6	67.0	75.1	76.7	72.0	61.8	51.9	41.0	(24)
	10%	DB	35.8	37.9	44.8	59.7	70.7	76.6	83.1	85.7	78.2	66.0	54.0	41.9	(25)
		MCWB	33.4	34.4	39.4	50.4	58.8	65.8	73.7	75.6	70.5	59.9	50.3	39.1	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.0	41.8	48.1	60.2	66.7	73.4	78.8	79.9	77.3	68.5	58.6	49.7	(27)
		MCDB	41.6	45.0	56.1	67.1	76.5	80.6	89.0	90.5	85.2	72.9	61.6	51.8	(28)
	2%	WB	36.8	38.8	45.0	56.3	64.3	71.0	77.3	78.4	75.3	65.4	55.2	45.0	(29)
		MCDB	38.4	42.0	49.9	64.4	72.6	77.2	86.4	87.8	82.1	69.2	57.8	47.3	(30)
	5%	WB	35.2	36.5	42.5	53.9	62.4	69.4	76.1	77.4	73.6	63.1	53.0	41.9	(31)
		MCDB	36.8	39.2	47.1	61.5	70.5	76.2	84.1	86.0	79.2	66.9	55.7	44.0	(32)
	10%	WB	33.8	34.8	40.1	51.5	60.6	67.8	74.8	76.4	71.7	61.2	50.7	39.4	(33)
		MCDB	35.1	37.2	44.1	58.1	68.3	74.0	81.4	83.9	76.8	65.1	53.5	41.3	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	8.9	10.9	14.1	19.0	19.0	16.2	14.0	15.2	15.0	15.7	13.2	8.9	(35)
		MCDBR	10.7	14.8	21.3	28.2	28.1	22.7	19.2	19.2	17.9	18.8	17.8	13.1	(36)
	5% WB	MCWBR	8.1	10.7	13.8	14.8	12.7	9.9	8.1	7.6	8.2	10.5	12.3	10.1	(37)
		MCWBR	10.3	13.8	19.4	22.6	21.3	16.9	17.0	17.6	15.3	15.0	15.9	12.8	(38)
(40) Clear-Sky Solar Irradiance	taub		0.387	0.425	0.492	0.531	0.545	0.533	0.542	0.522	0.460	0.426	0.409	0.396	(40)
		taud	1.957	1.859	1.822	1.830	1.857	1.965	2.010	2.065	2.207	2.192	2.134	2.041	(41)
	Ebn at Noon		233	242	240	240	240	243	239	240	249	243	228	220	(42)
		Edh at Noon	42	53	62	65	64	58	55	51	42	38	36	36	(43)
(44) All-Sky Solar Radiation	RadAvg		483	734	1066	1407	1583	1566	1379	1416	1162	897	589	413	(44)
	RadStd		43	63	100	144	162	114	195	166	88	75	46	45	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (14 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page